Dmytro Lamashevskyi

Senior Software Engineer | .NET, AWS & Applied AI

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Professional Summary

Senior Software Engineer with 6+ years of commercial experience building backend, distributed, and data-intensive systems. Hands-on background in .NET and AWS, spanning architecture, implementation, production diagnostics, business automation, and legacy modernization. At Honki Mode, reduced selected POS analytical update latency from approximately 1-2 minutes to 10-20 seconds and implemented approximately 300 MB CSV exports with bounded memory use. Applied AI work includes a commercial RAG feasibility prototype and personal projects using Amazon Bedrock and speech AI. Seeking a hands-on senior role with responsibility for architecture and product decisions.

Core Competencies

Architecture & Backend: C#, .NET, ASP.NET Core, REST APIs, distributed systems, microservices, system integration, solution design.
Cloud & Infrastructure: AWS Lambda, API Gateway, Cognito, S3, EC2, EKS, CloudWatch, Docker, serverless architecture, cost optimization.
Data & Messaging: PostgreSQL, SQL Server, ClickHouse, Redis, MongoDB, RabbitMQ, SQS, caching, background processing, analytics workloads.
Applied AI: Generative AI, RAG, AI agents, tool/function calling, Amazon Bedrock, MCP, local LLM inference, llama.cpp, whisper.cpp, Grok, Gemini.
Frontend & Desktop: React, TypeScript, WPF, Rust / Tauri, Python / Qt.
Engineering Practices: Architecture trade-offs, production diagnostics, performance optimization, automated testing, CI/CD, root-cause analysis.

Professional Experience

Full-Stack .NET Developer, Architecture Contributor **Jun 2024 - Present**
Honki Mode (Startup) *Kagawa, Japan*
POS Administration Platform *Production system*
Greenfield administration application for a distributed POS ecosystem

- After completing the chatbot feasibility work, joined the POS product in early development and built its administration application from scratch, integrating with the wider ecosystem through RabbitMQ from the outset.
- Designed an analytics layer using ClickHouse, RabbitMQ, and Redis, combining data from multiple services while caching relatively static reference data to reduce synchronous dependencies.
- Created a constrained JSON-to-SQL query layer that validates frontend-defined analytical requests and translates them into ClickHouse SQL without accepting arbitrary client SQL.
- Implemented cached background recalculation segmented by tenant and time range, with batching and locking for concurrent updates; reduced selected analytical update latency from approximately 1-2 minutes to 10-20 seconds.
- Built background CSV exports of approximately 300 MB across multiple services using pagination, batching, progress reporting, streaming generation, and direct S3 upload with bounded memory use.
- Introduced export reuse for identical filters and explicit stale-data handling, letting users download an existing file or force regeneration.
- Implemented an application layer around Amazon Cognito using access tokens, refresh tokens, and application cookies for renewal and session-consistency checks.
- Contributed to an infrastructure review and consolidation onto two EC2 worker nodes in the existing EKS environment; reduced estimated recurring cost by approximately $1,300 per month relative to the original monthly infrastructure estimate and introduced log and backup rotation policies.
- Proposed a project-specific versioned RabbitMQ contract approach, reviewed service boundaries, and recommended consolidation where independent deployment did not justify the operational complexity; applied tenant-aware design for future multi-client reuse.

Technologies: C#, .NET / ASP.NET Core, React, TypeScript, PostgreSQL, ClickHouse, Redis, RabbitMQ, AWS S3, Cognito, EC2, EKS, CloudWatch.

Professional Experience (continued)

Honki Mode **Jun 2024 - Present**

Current company - continued *Kagawa, Japan*

Store-Support GenAI Chatbot *Nov 2024 - Feb 2025 (approx. 2-3 months)*

Commercial feasibility prototype - end-to-end prototype design and implementation; not production

- Independently researched, designed, and built a separate React and .NET serverless prototype on AWS; the normal web path used API Gateway and Cognito, with Amazon Bedrock for model-backed responses and Lambda functions for backend and tool execution.
- Implemented a document RAG workflow using S3, an AWS-managed indexing/vectorization flow, an Amazon Aurora-backed vector knowledge layer, and a UI for updating knowledge and triggering re-indexing.
- Added tools for product search, orders, and carts; reproduced a cross-user data-isolation flaw caused by model-controlled identity parameters and identified trusted server-side identity as the required design boundary.
- The prototype was stopped after retrieval, security, and cost-to-impact evaluation. The proposed secure redesign was not implemented.

Technologies: React, .NET AWS Lambda, API Gateway, Cognito, Amazon Bedrock, S3, Amazon Aurora-backed vector storage, RAG, AI tool/function calling.

Barcode Master *Operational automation*

PDF processing, barcode generation, and printer integration

- Implemented PDF processing and barcode placement while preserving document geometry and print scale, preventing distortion that could invalidate barcode readability.
- Reduced the workflow requirement from two employees to one and cut manual processing time by more than half to approximately three hours or less.

AI Solutions Architect & Tech Consultant **Mar 2026 - Present**

Honki Mode (Startup) *Kagawa, Japan*

Client-facing solution engineering with Forward Deployed Engineer responsibilities: requirements discovery, business automation, system integration, and practical AI integration.

Forward Deploy - Payment Data Automation *Client-specific solution engineering*

Python / Qt finance and payment workflow

- Automated a monthly process that required at least approximately eight hours of manual CSV transfer, calculation, checking, and Excel preparation by converting source CSVs into normalized business data and ready-to-use payment workbooks.
- Added reusable account data, company-name conflict resolution, manual entry, and stored-data output as the tool evolved from CSV conversion into an internal application; estimated savings are approximately 1-2 person-days per month.

Forward Deploy - Custom ERP *Under active development*

Python / Django / JavaScript, dynamic pricing, and authenticated MCP integration

- Designed a custom ERP to replace fragmented Excel workflows with centralized product, material, pricing, quotation, order, contract, permission, and document data.
- Implemented administrator-defined parameters, dynamic product forms, and independent server-side pricing formulas so sales staff use shared rules without receiving editable calculation logic.
- Added authenticated MCP endpoints that let AI clients retrieve products and invoke the ERP's centralized pricing engine, keeping business rules in the ERP as the source of truth instead of duplicating them in AI clients.

Forward Deploy - APC-51 Legacy ERP Modernization *Under active development*

Read-only legacy integration and gradual workflow replacement

- Reverse-engineered a business-critical ERP and local database more than 20 years old, reconstructing schema relationships and operational data flow while identifying business-continuity risk around the aging host and undocumented behavior.
- Designed a new application with its own database and intentionally read-only synchronization from APC-51 after direct writes were judged too risky for the undocumented legacy system.
- Established a gradual modernization path from APC-51 through the new application database into modern order and warehouse workflows, reducing dependence on fragile Excel-based downstream processing without destabilizing the source system.

Professional Experience (continued)

Backend Developer **Feb 2023 - May 2024**
Devox Software *Poland / Remote*

- Developed backend services, REST APIs, and integrations for a UK university-admission platform supporting multi-university applications and centralized document verification.
- Created an internal library for dynamic model mapping and OpenAPI / Swagger documentation, reducing duplication between backend and frontend models and supporting frontend API client generation from current service contracts.
- Delivered microservice functionality in an AWS-backed environment using EC2, S3, Lambda, and CloudWatch.

Technologies: C#, ASP.NET Core, microservices, REST, Swagger / OpenAPI, AWS EC2, S3, Lambda, CloudWatch.

Desktop WPF Developer **Dec 2021 - Oct 2022**
Motorola Solutions *Krakow, Poland*

- Maintained a mission-critical Windows application used by mobile and patrol units across more than 40 U.S. counties, covering defect resolution, refactoring, performance, and L2 / L3 production support in an ITIL / ITSM and SLA-oriented environment.
- Diagnosed defects across a large, tightly coupled legacy codebase; in one representative case, traced a responsive-layout failure to an upstream configuration module that reapplied a fixed width after the layout engine recalculated the UI.
- Worked with Windows permissions, inter-process communication, mapping integrations, dynamically configured UI components, and synchronized operational data.
- Delivered changes through Azure Pipelines and formal enterprise quality processes backed by a regression suite of more than 3,000 tests.

Technologies: C#, .NET, WPF, Windows integration, Azure Pipelines, automated testing.

Earlier Commercial Experience

.NET Full-Stack Developer **Dec 2022 - Jan 2023**
CyverIO (Contract) *Remote*

- Delivered ASP.NET Core and Angular functionality for a penetration-testing management platform and contributed to architecture and refactoring decisions in a high-velocity environment with accumulated technical debt.

Backend Developer (Part-Time / Freelance) **Apr 2022 - Sep 2022**
Whimsy Games *Remote*

- Built backend and administration functionality for player and game data, including REST APIs and a lightweight authentication and request-validation flow using Redis and MongoDB in an AWS / Unity-oriented environment.

C# Developer **Feb 2020 - Oct 2021**
Sollers Consulting *Lublin, Poland*

- Customized Moody's CreditLens for banking clients, delivering SQL-based reports and customer-specific enterprise functionality.
- Optimized selected reporting workflows by approximately 30-40% through query redesign, caching, and parallel retrieval; clarified requirements with a Moody's manager and delivered internal knowledge-sharing sessions.

Selected Contract Engineering

Insurance Automation Platform *Automation Developer*
Concurrent processing, reliability, and diagnostics

- Designed a database-queued worker platform that processed up to 10 insurance applications concurrently through isolated Selenium flows and captured structured logs, context, and screenshots for external-site failures.

Scientific Desktop Application *Desktop Software Developer*
High-frequency visualization and hardware integration

- Built a behavioral-research application integrating National Instruments hardware and MATLAB workflows, with a dedicated rendering thread for synchronized multi-source visualization above 60 FPS without blocking the UI.

Selected Engineering & AI Projects

SchoolAI *Personal project - Creator, Full-Stack Developer, Product and Architecture Designer*
school-ai.jp

- Designed a serverless .NET 8 backend using AWS Lambda, API Gateway, Cognito, and RDS for an educational platform that analyzes student work and produces structured recommendations.
- Separated asynchronous AI workloads from interactive requests using SQS FIFO queues, a dead-letter queue, and worker Lambdas; delivered job updates through API Gateway WebSockets with connection state in DynamoDB.
- Integrated Amazon Bedrock for model inference and Systems Manager Parameter Store for configuration and secrets; built the React / TypeScript frontend with distinct administrative and student-facing workflows.

Technologies: C#, .NET 8, AWS Lambda, API Gateway, Cognito, RDS, SQS FIFO / DLQ, DynamoDB, WebSockets, Amazon Bedrock, React, TypeScript.

KaigiAI *Personal open-source project - Creator, Full-Stack / Systems Developer*
github.com/DmytroLamashevskyi/KaigiAI

- Built a multilingual meeting assistant in response to real workplace difficulties with conversation speed and translation; validated it during actual work meetings.
- Implemented a Tauri v2 desktop architecture with a Rust core and React / TypeScript UI, managing local whisper.cpp and llama.cpp sidecars through OpenAI-compatible localhost APIs.
- Combined WASAPI microphone and system-audio capture, Whisper speech recognition, optional ONNX-based diarization, SQLite history, and an LLM layer for context-aware translation and analysis of text and speech-derived content.
- Supported local inference and external providers including Grok and Gemini; used Grok in real meetings when the available PC was insufficient for satisfactory local LLM performance, while preserving a local-first mode for sensitive audio and transcripts.

Technologies: Rust, Tauri v2, React, TypeScript, whisper.cpp, llama.cpp, local LLMs, Grok, Gemini, SQLite, ONNX Runtime, WASAPI.

Orionyx Online *Personal project - Creator, Lead Developer, System Designer*
orionyx.online

- Designed and implemented a browser-based real-time space MMO strategy game with configurable universes, progression, fleets, combat, alliances, trade, quests, rankings, and administration tooling.
- Built a monorepo with a React SPA, Express API, PostgreSQL, Redis, HTTP, and Socket.IO communication, demonstrating end-to-end product and real-time system design outside the .NET ecosystem.

Technologies: React 19, TypeScript, Node.js, Express, PostgreSQL, Redis, Socket.IO, Docker.

Education

Master of Science in Computer Science, Zhejiang Normal University, China 2024
Bachelor of Science in Computer Science, Lublin University of Technology, Poland 2020
Bachelor of Science in Computer Engineering, Chernihiv National University of Technology, Ukraine 2018

Languages, Certification & Training

Languages: Ukrainian (Native), Russian (Native), Polish (Upper Intermediate), English (Upper Intermediate), Japanese (N4).
Certification: AWS Certified AI Practitioner – Amazon Web Services (Sep 2026 – Sep 2029).

Selected training and course-completion credentials: Claude Code in Action - Anthropic (Apr 2026); 5-Day AI Agents Intensive - Kaggle / Google (Dec 2025); AWS Cloud Quest: Generative AI Practitioner - AWS Training & Certification (Nov 2025); C# ASP.NET Core - OTUS, 156 academic hours (Jun 2023).